Customer Segmentation Analysis Using Retail Transaction Data

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AI Declaration
I would like to be transparent about the involvement of generative AI tools for this coursework.

Tools Used: I used Claude AI (Anthropic) and ChatGPT for debugging the code, organizing the report, and proofreading.

How it was used: The AI tools helped me refine the code, suggest ideas for enhancing the visualization, and provide feedback on how the report can be made easier to read. The analysis, models, understanding, and business recommendations are my own.

Declaration: I am aware that this work will be treated as mine, and I accept full responsibility for what is presented in this report.

Executive Summary

This report presents a customer segmentation analysis based on transactional retail data. The objective of such an analysis is to identify different groups of customers with similar purchasing behavior, which can be helpful for targeted marketing and better decision-making.

The data analysis incorporates four different data sets containing information on customers, transactions on a basket level, category-level transactions, and purchases made on a product level. Data preparation involved converting currency values into a numerical format, correcting erroneous data in category-level transactions, and creating behavioral data such as basket spending, recency, and time of day.

Customer behavioral and category features were used to create a wide feature matrix. Data preprocessing techniques such as winsorisation, logarithmic transformation, and robust scaling were used to optimize clustering results. Principal Component Analysis (PCA) has been used to reduce dimensionality while retaining most of the variance contained in the data.

Customer segments were identified by implementing a K-Means clustering algorithm. This algorithm is popular for customer segmentation because it can effectively divide large data sets into groups containing observations with similar characteristics. Based on inertia and silhouette scores, a five-cluster solution has been found to be appropriate. The identified clusters include different types of customers, such as grocery shoppers, convenience tobacco customers, cashpoint customers, drinks and tobacco customers, and lottery customers.

The results obtained can be helpful for targeted marketing strategies and better decision-making.

Customer Base Summary

The dataset provided by the client contains transactional information that describes the purchasing behavior of the customers in the retail store. The analysis integrates four datasets, including the customer identification data, transactional basket data, category spending data, and line-item purchase data. These datasets offer a comprehensive overview of the purchasing behavior of the customers.

The dataset contains information regarding 3,000 customers, including over 195,000 basket transactions and one million line-item purchases. The large number of transactions in the provided dataset can be seen as a strong indicator of the presence of patterns in the purchasing behavior of the customers.

To gain better insight into the overall characteristics of the customers, some behavioral features have been developed from the transactional data provided in the dataset. These features of the customers' purchasing behavior include the purchase frequency, total basket spending, average basket spend, average basket quantity, and average number of product categories purchased per basket.

The exploratory analysis of the features of the purchasing behavior of the customers indicates that the purchasing behavior of the customers varies significantly. Some customers show higher purchase frequency with higher basket spend, while others show lower purchase frequency with lower basket spend, purchasing products in the categories in which they have previously made purchases.

The overall analysis of the features of the purchasing behavior of the customers indicates that the customers of the retail store do not belong to a homogeneous group. Instead, the customers show different purchasing

behavior, which can be seen as a strong indicator that the provided dataset can be effectively utilized for the purpose of segmentation analysis.

The next part of the analysis involves the development of features that can effectively differentiate the purchasing behavior of the customers to allow for the clustering of the customers into different groups.

Feature Engineering and Description

For the clustering to be effective, the variables must account for the differences in the purchasing behavior of the customers. As such, both behavioral features and product preference features were developed from the available data.

The behavioral features were developed from the basket transactional data set. These features included purchase frequency, total basket spending, average basket spend, average basket quantity, and the average number of product categories per basket.

The recency of the purchase was also developed by determining the number of days that had elapsed since the last purchase by the customer. Temporal behavioral features were also developed by determining the time of day and the weekend of purchase. The time of day was categorized into morning, afternoons, and evenings, and the proportion of purchases in each of the time segments was determined for each customer.

The proportion of purchases that took place over the weekends was also determined to account for the differences in the purchasing behavior that may exist over the weekends as opposed to the other days of the week. In addition to the behavioral features, the product preference features were developed from the category spending data set. The proportion of the total category spending to the category spending of the customer was determined to account for the differences in the purchasing behavior of the customers. To reduce the redundancy of the features developed, some of the features that were highly correlated were removed from the set of features.

The derived features included the spend per visit, which was determined by dividing the total basket spending by the purchase frequency. The final set of features that was developed for the clustering analysis comprised twenty-five variables.

Segmentation Methodology

Prior to clustering, several preprocessing steps were carried out to enhance model performance and limit the impact of extreme values. Winsorisation was used on several magnitude-based features such as basket spend, basket quantity, and recency. This limits the impact of extreme values that could otherwise skew clustering results.

Logarithmic transformations were carried out on several magnitude-based features. This limits the impact of extreme values on these features.

All features were subsequently rescaled using the RobustScaler method. This method rescales data using median and interquartile range values instead of mean and standard deviation values. This method is particularly beneficial for data that contains outliers and limits their impact during clustering.

Principal Component Analysis (PCA) was carried out on the data to reduce dimensionality while retaining a high percentage of variance. It was found that six principal components retained 81% of the variance within the data.

Customer segments were determined by using a K-Means clustering algorithm. This algorithm groups data into a set of clusters such that customers within a particular cluster are more similar to each other than they are to customers in other clusters. This algorithm is particularly beneficial because it is an efficient algorithm and can be used on large data sets.

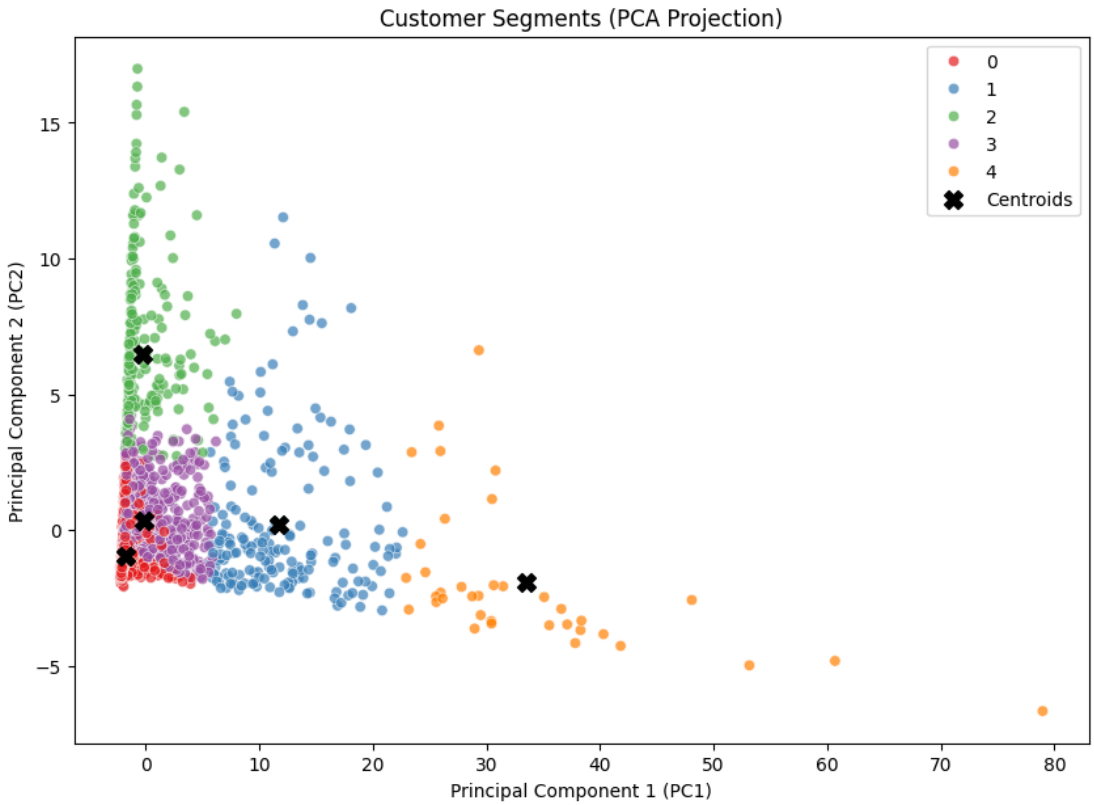

Figure 1: PCA projection of customer segments identified using the K-Means clustering algorithm.

The PCA projection provides a visualization of the results obtained by clustering, considering the reduced feature space and the results obtained by considering the first two components. Each point on the plot represents a customer, and they are represented by different colors depending on their respective clusters. The black dots on the plot indicate the centroids of each cluster. The plot indicates that different clusters are in different regions of the feature space, implying that there are significant differences between customers. Despite some overlapping values, it can be concluded that the separation between different clusters is significant and indicates a valid clustering model.

To determine the appropriate value of k for each model, different values of k were used to compute inertia and silhouette scores. Figure 2 presents the results obtained by using different values of k for inertia and silhouette scores.

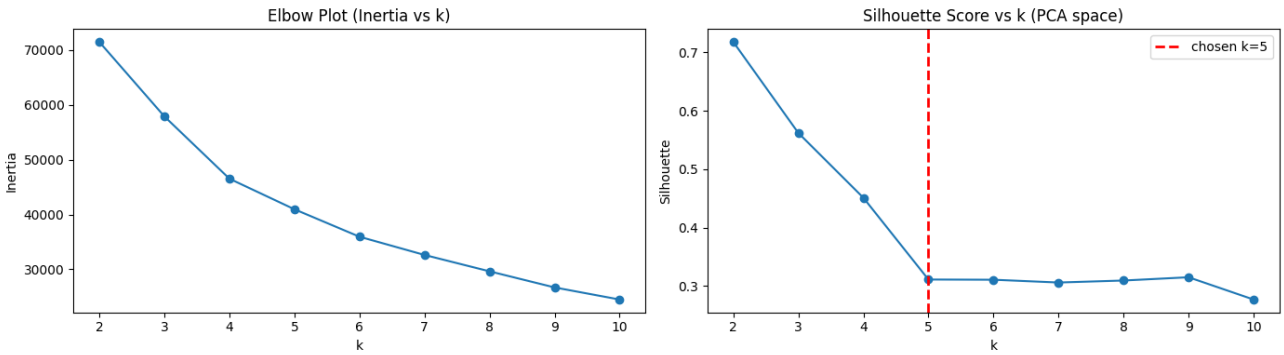

Figure 2: Elbow + Silhouette Plot

As shown in Figure 2, inertia is consistently declining as the number of clusters increases, and the silhouette score is relatively stable at five clusters. This implies that a model with five clusters is a reasonable choice. On the basis of the above analysis, a five-cluster solution is chosen because it offers a reasonable trade-off between cluster separateness and model simplicity.

Results

5.1 Segment Distribution

The clustering analysis identified five distinct customer segments with meaningful behavioral differences. These segments vary in terms of spending behavior, purchase frequency, product category preferences, and shopping patterns.

The distribution of customers across the identified segments is illustrated in Figure 3.

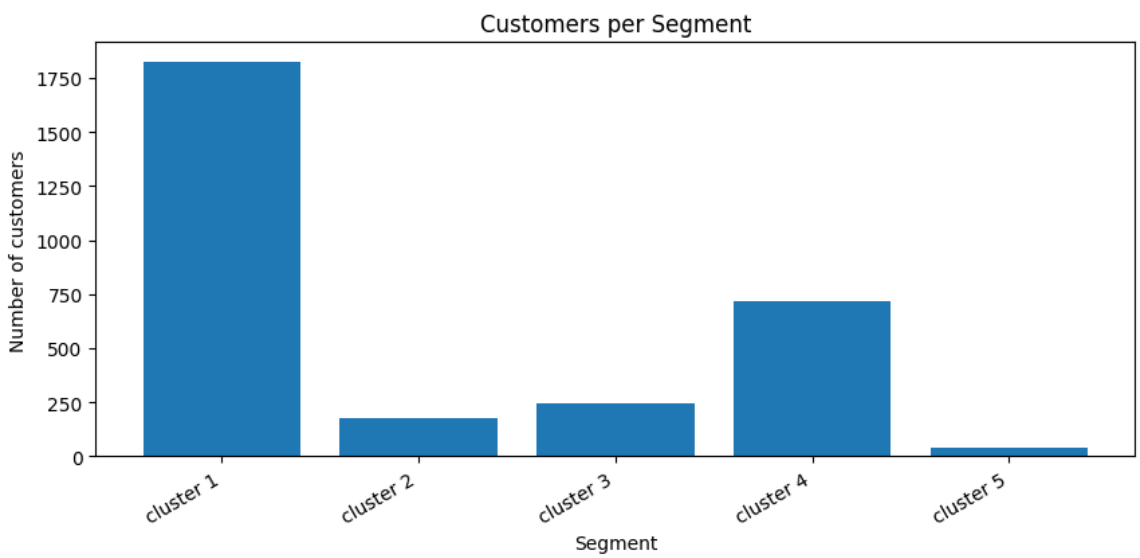

Figure 3: Distribution of customers across the five identified customer segments.

As can be seen from Figure 3, the number of customers in the different clusters varies. However, the highest number of customers can be found in Cluster 1. This suggests that the larger proportion of the customers have similar patterns of purchasing behavior. However, the smaller number of customers in the other clusters implies that these customers have specialized purchasing behavior. Other smaller clusters, such as Cluster 5, have smaller numbers of customers. These smaller clusters represent specialized purchasing patterns that are significantly different from the purchasing behavior of the larger number of customers. The identification of these specialized purchasing patterns is important for retailers since it enables the formulation of marketing strategies for the different customers.

5.2 Statistical Comparison of Segments

The largest segment has the greatest number of customers, the highest average basket spend, and the greatest number of product categories per basket.

The other segments show different patterns of specialization in purchasing behavior. One segment shows significantly higher spending on tobacco and lottery products, suggesting that these customers primarily make convenience purchases.

Another segment has significantly higher usage of the cashpoint services in the store, implying that some customers primarily make financial transactions in the store as opposed to purchasing grocery products. The box plot analysis has verified that the behavioral variables such as basket spend, basket quantity, category diversity, and purchase recency show significant variation among the segments, implying that the clustering model has successfully identified different behavioral patterns among the customers. Figure 4: Differences in Key Behavioral Variables among Clusters

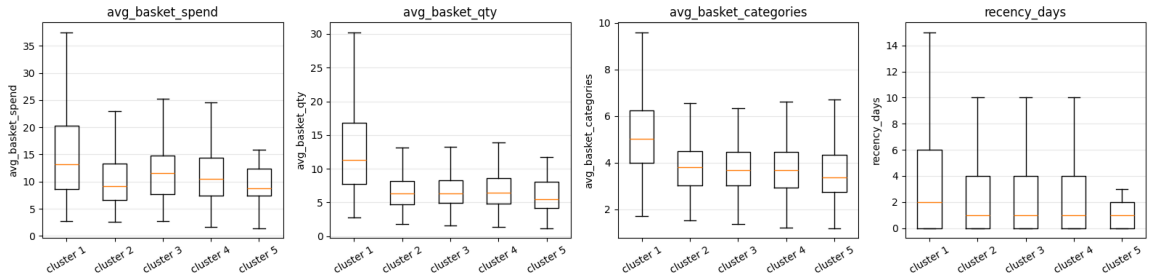

Figure 4: Comparison of behavioral metrics across customer segments.

The box plots in Figure 4 reveal clear distinctions in behavioral characteristics between clusters. For instance, Cluster 1 indicates that it has the highest median basket spend and basket quantity, implying that this cluster of customers tends to make larger purchases in terms of variety and quantity. On the other hand, other clusters like Cluster 2 and Cluster 5 indicate lower basket spends and quantities, implying that these customers tend to make smaller purchases for convenience. The observed behavioral characteristics reveal that the clustering model has been successful in grouping customers with different purchasing behaviors into distinct clusters. These behavioral characteristics form the basis upon which effective customer segment profiles can be developed, as discussed in the following pen portraits.

Segment	Relative Size	Basket Spend	Basket Quantity	Product Diversity	Recency	Key Behavior
Core Grocery Shoppers	Largest	High	High	High	Moderate	Full grocery shopping trips
Tobacco Convenience Buyers	Small	Low–Moderate	Low	Low	Recent	Convenience tobacco purchases
Cashpoint Users	Small	Low	Low	Very Low	Frequent visits	Cashpoint transactions
Drinks & Tobacco Shoppers	Medium	Moderate	Moderate	Moderate	Moderate	Beverage and tobacco purchases
Lottery-Focused Visitors	Very small	Low	Very low	Very low	Recent	Lottery purchases

Table summarizes the key behavioral characteristics that distinguish each customer segment identified by the clustering analysis.

Pen Portraits of Clusters

The core grocery shoppers form the largest segment of the target customer base. These customers have the highest average basket spend, as well as the highest number of different product categories that they purchase. The purchases of the core grocery shoppers are primarily in staple grocery products such as fruit and vegetables, dairy products, and other packaged grocery products.

The Tobacco Convenience Buyers form the second segment of the target customer base. These customers have relatively high average spend in the tobacco category in comparison to other categories. The Tobacco Convenience Buyers purchase fewer grocery products, primarily for convenience purchases. The Cashpoint Users form the third segment of the target customer base. These customers have the highest usage of the cashpoint services provided by the store. The purchases of the Cashpoint Users primarily consist of convenience purchases, as opposed to grocery shopping.

The Drinks and Tobacco Shoppers form the fourth segment of the target customer base. These customers have strong preferences for drinks and tobacco products, primarily purchasing convenience products. The Lottery Focused Visitors form the smallest segment of the target customer base but have extremely high spending in the lottery category in comparison to other categories. The purchases of the Lottery Focused Visitors primarily consist of convenience purchases as opposed to grocery shopping.

Summary and Business Recommendations

This analysis offers important insights that can be used to improve marketing strategies and operational activities. The Core Grocery Shopper is the most important segment in the market because it has the highest level of spending and product purchasing. Loyalty programs and personalized marketing programs can be used to retain this important customer base and increase basket sizes and purchase frequencies.

Tobacco and beverage segments are likely to respond well to convenience marketing strategies, and placing such products near the entrance or checkout counters can be a great marketing move for the store. The customer segments that make frequent use of the store's cashpoint facility are a potential area that can be explored for cross-selling activities. Impulse product sales can be maximized by placing such products near the cashpoint facility, where these customers are likely to make impulse purchases while transacting at the facility. The lottery segments can be targeted for marketing activities that promote impulse sales in conjunction with lottery sales. In conclusion, the segmentation analysis has provided important insights into how customer behavioral data can be used to identify important segments and develop targeted marketing strategies and improve customer engagement in a retail environment.

Technical Implementation

The analysis was performed using the Python language in a structured notebook environment. The entire workflow involved a series of steps that included loading the data, cleaning the data, feature engineering, preprocessing, dimension reduction, clustering, and interpretation of the results. The code has been organized into logical segments to make it easier to read and maintain. The variables have been named in such a manner that it is easy to understand the purpose of the features in the code. The code also contains comments that explain the important steps in the analysis.

The results of the segmentation analysis were saved in a CSV file that contains the customer IDs along with the segment information. This makes it easier to incorporate the results of the segmentation analysis into business applications. The entire workflow of the segmentation analysis has been designed in such a manner that it can be reproduced in the future as well, as new customer information becomes available.